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1. SYSTEM OVERVIEW

APWRS is a drought early warning system. It ingests 6 categories of environmental data and uses 3 prediction methods (XGBoost, LSTM neural networks, physics-based heuristics) to forecast drought risk 7 and 30 days ahead.

Why 6 Metric Categories?

Agriculture is a multi-dimensional system. No single metric tells the full story:

Category	What It Captures	Why It's Essential
Soil	Water storage capacity & availability	The "fuel tank" for crops
Climate	Weather inputs (rain, temperature, etc.)	The "fuel pump"
Drought Indices	Standardized historical drought context	The "baseline comparison"
Agricultural	Crop-specific water demand & stress	The "engine load"
Remote Sensing	Actual vegetation health from space	The "objective measurement"
Hydrology	Overall water resources balance	The "regional water budget"

2. METRIC GROUP 1: SOIL METRICS

2.1 moisture_content_percent

Aspect	Detail
Why chosen	Direct measure of water available to plants. The single most important soil drought indicator.
What it means	Percentage of soil volume occupied by water. 0% = bone dry, 100% = fully saturated.
How measured	Gravimetric method (weigh wet soil, dry at 105°C, re-weigh) or time-domain reflectometry (TDR) sensors, or satellite microwave sensors.
Effect on drought	Below ~30% = plants begin to stress. Below ~15% = permanent wilting.
Calculation	$\text{moisture_pct} = (\text{weight_wet} - \text{weight_dry}) / \text{volume_soil} \times 100$
AI usage	Primary input for XGBoost (latest value + 7-day trend). LSTM uses it as the target variable to predict (soil moisture 7/30 days out). Heuristic: $\text{soil_risk} = (1 - \text{moisture}/\text{field_capacity}) \times 100$.

2.2 sand_ratio, clay_ratio, silt_ratio

Aspect	Detail
Why chosen	Soil texture determines water holding capacity, drainage, and infiltration. Sand drains fast, clay holds water, silt is intermediate.
What it means	Percentage of each particle size in the soil. Must sum to 100%.
How measured	Hydrometer method or laser diffraction.
Effect on drought	Sandy soil = low water holding → faster drought onset. Clay soil = higher holding but slower recharge.
AI usage	Used to calculate field capacity & wilting point via Saxton-Rawls pedotransfer functions (see Section 11). These are then used to normalize soil moisture into risk percentages.

2.3 root_zone_depth_mm

Aspect	Detail
Why chosen	Defines the volume of soil from which plants can extract water. Deep roots access more stored water.
What it means	Depth in mm of the effective root zone (typically 300-1200mm for field crops).
How measured	Soil pit excavation, root coring, or estimation by crop type.
Effect on drought	Deeper root zone = more total available water = longer drought survival.
Calculation	$\text{Available Water Capacity (mm)} = (\text{FC} - \text{WP}) / 100 \times \text{root_zone_depth_mm}$
AI usage	Determines <code>available_water_capacity_mm</code> which scales all risk calculations.

2.4 organic_matter_percent

Aspect	Detail
Why chosen	Organic matter improves water retention (can hold 20x its weight in water).
What it means	Percentage of soil mass that is decomposed plant/animal material.
How measured	Loss on ignition (burn at 550°C, measure weight loss) or Walkley-Black wet oxidation.
Effect on drought	Higher OM = better structure, more infiltration, more water storage = reduced drought impact.

AI usage	Influences field capacity calculation in Saxton-Rawls.
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2.5 infiltration_rate_mmhr

Aspect	Detail
Why chosen	Determines how quickly rainwater enters the soil vs. running off.
What it means	Millimeters of water that can enter per hour. Sandy soil: 50+ mm/hr. Clay soil: 1-10 mm/hr.
How measured	Double-ring infiltrometer or rainfall simulator.
Effect on drought	Low infiltration = more runoff = less water stored for dry periods.
AI usage	Used indirectly in the water balance for runoff estimation.

2.6 field_capacity_percent & wilting_point_percent

Aspect	Detail
Why chosen	These two thresholds define the **plant-available water window** . FC = upper limit (gravity has drained), WP = lower limit (plants can no longer extract).
What it means	FC: ~30-40% for clay, ~10-15% for sand. WP: ~15-20% for clay, ~4-6% for sand.
How measured	Pressure plate apparatus, or **estimated** via Saxton-Rawls from texture: `FC = 20 + 0.3×clay + 0.1×silt - 0.05×sand` / `WP = 8 + 0.2×clay + 0.05×silt - 0.03×sand`
Effect on drought	When moisture falls below FC = stress begins. At WP = plant death. The danger zone is FC → WP.
AI usage	**Critical** : All three models normalize soil moisture against field capacity to compute risk. `risk = (1 - moisture/FC) × 100`.

2.7 salinity_ece_dsm

Aspect	Detail
Why chosen	High salinity causes "physiological drought" — water is present but plants cannot absorb it.
What it means	Electrical conductivity in deciSiemens per meter. 0-2 = normal, 2-4 = slight, 4-8 = moderate, 8+ = severe.
How measured	EC meter on soil-water slurry.
Effect on drought	Saline soils reduce effective water availability, increasing drought sensitivity.
AI usage	Secondary input. Not directly in current models but affects interpretation.

2.8 ph_level

Aspect	Detail
Why chosen	pH controls nutrient availability. Extreme pH stresses crops even with adequate water.
What it means	1-14 scale. 6.0-7.5 = optimal for most crops.

How measured	pH meter on soil-water suspension.
Effect on drought	Indirect — stressed crops from poor pH are more vulnerable to drought impact.
AI usage	Informational secondary metric.

3. METRIC GROUP 2: CLIMATE METRICS

3.1 rainfall_mm

Aspect	Detail
Why chosen	Primary water input to the system. Without rain, drought is inevitable.
What it means	Depth of precipitation in millimeters over a 24-hour period.
How measured	Tipping-bucket rain gauge, weighing gauge, or satellite (TRMM/GPM).
Effect on drought	Direct inverse: more rain = less drought. But timing matters — 100mm in one day vs. spread over 2 weeks have very different impacts.
AI usage	**Critical** . XGBoost uses `rainfall_30d_mm` (sum over 30 days) and `rainfall_7d_mm` (sum over 7 days). Also computes `days_since_last_rain` (>1mm threshold). LSTM includes it as one of 10 input features. Heuristics use it for water balance.

3.2 seasonal_rainfall_variability

Aspect	Detail
Why chosen	Captures the distribution pattern — same total rain but different arrival timing.
What it means	Classification like "Normal", "Erratic", "Late onset", "Early cessation".
How measured	Historical comparison of rainfall timing vs. long-term averages.
Effect on drought	Late rains delay planting. Early cessation cuts short the growing season. Both cause crop water stress even if total rainfall is normal.
AI usage	Informational / metadata field.

3.3 temperature_max_c, temperature_min_c, temperature_mean_c

Aspect	Detail
Why chosen	Temperature controls evapotranspiration rate (the "thirst" of the atmosphere). Higher = more water loss.
What it means	Daily maximum, minimum, and mean air temperature in °C at 2m height.
How measured	Thermometer in a Stevenson screen (standard weather station).
Effect on drought	Higher temps = higher ET0 = faster soil drying. Daily max > 40°C causes direct heat stress on crops.
AI usage	**Critical** . XGBoost computes `temp_anomaly_c = current_temp - 30_day_avg_temp`. This captures heat waves. LSTM uses all three. Mean temp is used to calculate ET0 via Penman-Monteith.

3.4 relative_humidity_percent

Aspect	Detail
Why chosen	Humidity controls the drying power of air. Low humidity = faster evaporation.
What it means	Percentage of water vapor in air relative to saturation. 20% = very dry, 80% = humid.
How measured	Hygrometer or psychrometer (wet/dry bulb thermometer).
Effect on drought	Low humidity accelerates soil and plant water loss. Combined with high wind and temp = extreme ET.
AI usage	Input to Penman-Monteith ET0 calculation. Also an LSTM feature.

3.5 wind_speed_ms

Aspect	Detail
Why chosen	Wind removes the humid boundary layer near leaves, accelerating transpiration.
What it means	Average wind speed in meters per second at 2m height.
How measured	Anemometer at 2m or 10m standard height (then adjusted).
Effect on drought	Strong wind = more ET = faster soil drying. Especially dangerous when combined with heat.
AI usage	Input to Penman-Monteith ET0 calculation and LSTM feature.

3.6 solar_radiation_mjm2day

Aspect	Detail
Why chosen	Solar radiation is the energy source for evaporation. More sun = more water loss.
What it means	Megajoules per square meter per day. ~5-10 MJ/m2/day in winter, ~25-30 in summer.
How measured	Pyranometer (measures incoming shortwave radiation).
Effect on drought	Higher radiation = higher potential ET = faster depletion of soil moisture.
AI usage	Converted to W/m2 ($SR_{wm2} = SR_{mjm2day} / 0.0864$) and used in Penman-Monteith. Also an LSTM feature.

3.7 evapotranspiration_et0_mmday & evapotranspiration_etc_mmday

Aspect	Detail
Why chosen	ET0 is the "atmospheric demand" — how much water would evaporate from a reference grass surface. ETc is the actual crop-specific demand.
What it means	ET0: 3-7 mm/day typical. ETc: varies by crop coefficient ($Kc \times ET0$).
How measured	Calculated (not directly measured): Penman-Monteith FAO-56 equation (see Section 11). Lysimeters can directly measure.
Calculation	$ET0 = (0.408 \times \Delta \times Rn + \gamma \times (900 / (T + 273)) \times u2 \times (es - ea)) / (\Delta + \gamma \times (1 + 0.34 \times u2))$ then $ETc = ET0 \times Kc$

Effect on drought	When $ET_c > \text{rainfall} + \text{available soil water}$ = deficit = drought.
AI usage	**Critical** . XGBoost uses <code>`etc_7d_avg_mm`</code> and <code>`etc_30d_avg_mm`</code> as drought drivers. High ET = high risk. LSTM includes <code>`etc_mm`</code> as a feature.

4. METRIC GROUP 3: DROUGHT INDICES

4.1 spi_1month, spi_3month, spi_12month

Aspect	Detail
Why chosen	SPI (Standardized Precipitation Index) is the World Meteorological Organization's recommended drought index. It compares current rainfall to historical norms at different timescales.
What it means	A unitless value from -3 to +3. Negative = dry, Positive = wet. -1 to -1.5 = Moderate drought, -1.5 to -2 = Severe, < -2 = Extreme.
How calculated	<code>`SPI = (current_rainfall - historical_mean) / historical_std_dev`</code>
Why 3 timescales	SPI-1 captures short-term (soil moisture), SPI-3 captures seasonal (crop stress), SPI-12 captures long-term (groundwater/reservoir).
Effect on drought	SPI < -1: agricultural drought likely. SPI < -2: severe, widespread crop failure.
AI usage	XGBoost includes SPI in its heuristic target calculation (lower SPI = higher penalty). Also used in the heuristic fallback risk formula. Auto-calculated when climate data is ingested.

4.2 spei_1month, spei_3month, spei_12month

Aspect	Detail
Why chosen	SPEI adds the effect of temperature (via ET) to SPI. Climate change makes this increasingly important — even with normal rain, higher temps = more drought.
What it means	Same scale as SPI (-3 to +3) but incorporates water demand (ET) not just supply (rainfall).
How calculated	<code>`SPEI = (P - PET_climatic_balance - mean) / std_dev`</code> using a log-logistic probability distribution.
Effect on drought	More sensitive to warming than SPI. Can show "drought" even with normal rainfall if temperatures are high.
AI usage	Used alongside SPI in heuristic risk calculations. Provides temperature-aware drought context.

4.3 pdsi_value

Aspect	Detail
Why chosen	Palmer Drought Severity Index incorporates soil moisture storage, runoff, and ET in a water balance model. More comprehensive than SPI.
What it means	-4 = Extreme drought, -3 = Severe, -2 = Moderate, -1 = Mild, 0 = Normal.
How calculated	Complex water balance model using P, ET, soil moisture storage, and runoff with climate coefficients calibrated to local conditions.
Effect on drought	PDSI < -3 indicates serious, prolonged drought requiring intervention.
AI usage	Included in the prediction frame as contextual data.

4.4 drought_severity_class

Aspect	Detail
Why chosen	A human-readable classification that translates numerical indices into actionable categories.
What it means	None → Abnormally Dry → Moderate → Severe → Extreme.
How determined	Based on SPI/SPEI/PDSI thresholds combined.
AI usage	Output of the system (generated by prediction). Also useful for training classification models.

5. METRIC GROUP 4: AGRICULTURAL METRICS

5.1 growth_stage & crop_coefficient_kc

Aspect	Detail
Why chosen	Water demand changes dramatically through the growing season. The same crop needs 3x more water at flowering than at germination.
What it means	Kc is a multiplier applied to ET ₀ to get ET _c . Kc = 0.3-0.4 (initial), 1.0-1.2 (mid-season), 0.3-0.5 (late).
How determined	FAO-56 tables provide Kc curves for each crop by growth stage. Or measured by lysimeter.
Effect on drought	Mid-season (flowering/grain fill) is most sensitive to water stress. A drought during this phase causes maximum yield loss.
AI usage	Kc is used to compute $ET_c = ET_0 \times Kc$, which drives the water balance model. The growth stage helps contextualize risk severity.

5.2 crop_water_requirement_mmday (CWR)

Aspect	Detail
Why chosen	The actual daily water needed by the crop to avoid stress. Direct comparison against available water.
What it means	Typically 3-8 mm/day for most crops at peak season.
How calculated	$CWR = ET_0 \times Kc$ or more specifically $CWR = ET_c - \text{effective_rainfall}$
Effect on drought	When $CWR > \text{water supply (rain + irrigation + soil storage)}$, deficit = drought stress.
AI usage	Used in water balance computation. Gap between CWR and supply drives risk.

5.3 yield_reduction_factor

Aspect	Detail
Why chosen	Quantifies the crop-specific sensitivity to water stress. Some crops (maize) are very sensitive, others (olive) are drought-tolerant.
What it means	FAO Ky value: 0 (no sensitivity) to 1.5+ (very sensitive). Ky = 1.25 for maize, 1.0 for wheat, 0.85 for barley.

How	Determined from FAO-66 tables (yield response to water).
Effect on drought	Higher Ky = more yield loss per unit of water deficit.
AI usage	Helps contextualize risk — same moisture deficit causes more damage to high-Ky crops.

5.4 irrigation_method, irrigation_efficiency_percent, water_applied_mm

Aspect	Detail
Why chosen	Irrigation is the primary human intervention to mitigate drought. Method and efficiency determine how much water actually reaches the crop.
What it means	Drip: 90% efficient. Flood: 40-60%. Center pivot: 75-85%. Water applied: actual volume delivered.
How measured	Flow meters, or estimated from pump hours × flow rate.
Effect on drought	Efficient irrigation can compensate for rainfall deficits. But if irrigation supply is limited, drought hits harder.
AI usage	Water balance includes `irrigation_mm` as an input. The pipeline can model "what-if" irrigation scenarios.

5.5 leaf_temperature_c & stomatal_conductance

Aspect	Detail
Why chosen	Plant stress indicators that precede visible wilting. Canopy temperature > air temperature = stomatal closure = stress.
What it means	Stomatal conductance (mol CO ₂ /m ² /s): 0.05-0.5 normal, <0.05 = stressed. Leaf temperature: typically within 2-3°C of air temp; >5°C above = heat stress.
How measured	Infrared thermometer (leaf temp), porometer (stomatal conductance).
Effect on drought	Early warning — stomata close to conserve water, reducing photosynthesis. Leaf temp rises because evaporative cooling stops.
AI usage	Advanced warning features. Not currently primary model inputs but validate risk levels.

6. METRIC GROUP 5: REMOTE SENSING METRICS

6.1 ndvi (Normalized Difference Vegetation Index)

Aspect	Detail
Why chosen	NDVI is the gold standard satellite vegetation health metric. Measures greenness/chlorophyll activity.
What it means	-1 to 1. Barren: 0-0.1. Sparse: 0.1-0.3. Moderate: 0.3-0.5. Dense/healthy: 0.5-0.9.
How measured	Satellite: $\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$. NIR reflects healthy vegetation, Red is absorbed by chlorophyll. Sentinel-2, MODIS, Landsat.
Effect on drought	NDVI declining over time = vegetation stress = drought impact. NDVI < 0.3 in growing season = serious concern.
AI usage	**Critical** . XGBoost uses `ndvi` (latest value) and `ndvi_trend` (linear fit over 14 days). Declining NDVI = vegetation stress = higher risk. LSTM includes NDVI as one of 10 features. Heuristic: $\text{ndvi_penalty} = \max(0, (0.8 - \text{NDVI}) / 0.8 \times 20)$.

6.2 ndwi (Normalized Difference Water Index)

Aspect	Detail
Why chosen	NDWI detects water content in vegetation and open water bodies. More sensitive to plant water status than NDVI.
What it means	$(\text{Green} - \text{NIR}) / (\text{Green} + \text{NIR})$. Values > 0 = water, < 0 = dry.
How measured	Satellite sensors with green and NIR bands (e.g., Sentinel-2).
Effect on drought	NDWI drops before NDVI in water stress — earlier warning signal.
AI usage	Informational / cross-validation metric. Not a primary model input.

6.3 land_surface_temperature_c (LST)

Aspect	Detail
Why chosen	LST measures actual ground/vegetation temperature from space. Crop water stress causes temperature to rise.
What it means	Can be 10-20°C hotter than air temperature in dry conditions. Healthy vegetation stays cooler via transpiration.
How measured	Thermal infrared bands (e.g., MODIS band 31/32, Landsat TIRS, Sentinel-3 SLSTR). Split-window algorithm.
Effect on drought	Rising LST + constant or falling NDVI = severe crop water stress. LST anomaly $> 5^{\circ}\text{C}$ = alert.
AI usage	Critical . XGBoost uses <code>lst_c</code> (latest value). LSTM includes it as a feature. High LST = high ET = high risk.

6.4 satellite_soil_moisture_percent

Aspect	Detail
Why chosen	Satellite-based soil moisture provides spatially continuous measurements without field sensors.
What it means	Percentage of soil water content in the top 5 cm (SMAP) or deeper.
How measured	SMAP (L-band microwave, 0-5cm), Sentinel-1 (C-band SAR), or MODIS thermal inertia.
Effect on drought	Direct soil moisture measurement complements ground data. Gradient between satellite and ground = insight.
AI usage	Used as a fallback when ground-based <code>soil_moisture_pct</code> is missing. The pipeline fills gaps with satellite data.

6.5 satellite_source

Aspect	Detail
Why chosen	Different satellites have different resolutions, revisit times, and accuracy. Knowing the source helps assess data reliability.
What it means	SMAP (36km), Sentinel-1 (10m, 6-day revisit), MODIS (250m, daily), Landsat (30m, 8-day).
AI usage	Metadata for data quality weighting.

6.6 vegetation_condition_index (VCI)

Aspect	Detail
Why chosen	VCI normalizes NDVI against historical range — shows how current vegetation compares to "normal" for that time of year.
What it means	0-100%. <35% = drought, 35-50% = moderate, >50% = normal.
How calculated	$VCI = (NDVI_current - NDVI_min_historical) / (NDVI_max_historical - NDVI_min_historical) \times 100$
Effect on drought	VCI < 35% even with normal rainfall indicates a problem (maybe pest, disease, or soil issue).
AI usage	Included as contextual data in the prediction frame.

6.7 evapotranspiration_sebal_mmday

Aspect	Detail
Why chosen	SEBAL/METRIC are satellite-based energy balance models that map actual ET across large areas.
What it means	Actual ET in mm/day derived from satellite thermal imagery, not just potential/reference.
How measured	SEBAL: Surface Energy Balance Algorithm for Land. Uses LST, NDVI, and weather data to solve the energy balance.
Effect on drought	Actual ET < potential ET = crops are conserving water (stomatal closure) = stress.
AI usage	Cross-validation with ground-based ETc.

7. METRIC GROUP 6: HYDROLOGY METRICS

7.1 precipitation_mm & evapotranspiration_mm

Aspect	Detail
Why chosen	The two sides of the water balance equation. P - ET = net gain or loss.
What it means	Direct hydrological measurements (may differ from climate rainfall/ET because they're collected from hydrometeorological stations).
Effect on drought	Prolonged P < ET = accumulating water deficit = hydrological drought.
AI usage	Used in water balance and as cross-check on climate metrics.

7.2 groundwater_depth_m

Aspect	Detail
Why chosen	Groundwater is the reserve tank. Declining water table = long-term drought signal.
What it means	Depth in meters from surface to water table. Rising = good (recharge), falling = bad (depletion).
How measured	Piezometer / observation well.

Effect on drought	Shallow water tables (< 5m) can support crops via capillary rise. Deep tables (> 20m) = no benefit. Rapid decline = chronic overdraft.
AI usage	Contextual — informs long-term water availability.

7.3 runoff_mm & river_flow_m3s

Aspect	Detail
Why chosen	Runoff and river flow determine water available for reservoir storage and irrigation diversions.
What it means	Runoff: water flowing over/through soil into streams. River flow: actual discharge in cubic meters per second.
How measured	Runoff: weirs/flumes or model estimation. River flow: stage-discharge rating curve at gauging station.
Effect on drought	Low river flow = reduced irrigation supply = increased drought impact on agriculture.
AI usage	Used to model water availability for irrigation in the water balance.

7.4 reservoir_storage_m3 & irrigation_supply_available_m3

Aspect	Detail
Why chosen	Reservoirs store water from wet season for dry season. Storage level directly indicates drought resilience.
What it means	Current volume of water in reservoirs. % of total capacity.
How measured	Stage (water level) to volume conversion using bathymetric curves.
Effect on drought	Low reservoir storage = limited irrigation capacity = high drought risk even if rain is normal.
AI usage	Used to constrain irrigation supply in water balance scenarios.

7.5 soil_water_deficit_index_mm

Aspect	Detail
Why chosen	Direct supply-demand gap: how much more water is needed vs. available.
What it means	$\text{Deficit} = \text{ETc} - (\text{rainfall} + \text{irrigation} + \text{available_soil_water})$. Positive = shortage.
Effect on drought	Deficit > 50mm cumulative = serious crop water stress.
AI usage	Direct drought indicator. Included in prediction frame.

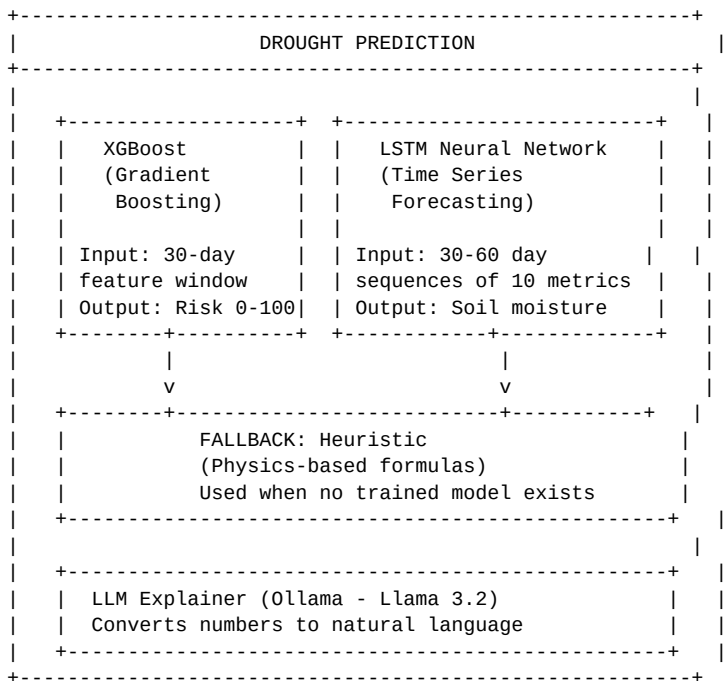
7.6 water_balance_percent

Aspect	Detail
Why chosen	Single-number summary of the regional water situation. Easy to communicate.
What it means	$(\text{supply} / \text{demand}) \times 100$. 100% = balanced, < 100% = deficit, > 100% = surplus.

Effect on drought	< 70% = drought conditions requiring monitoring. < 40% = crisis.
AI usage	Output-like metric that helps validate risk scores.

8. HOW AI PREDICTS DROUGHT

8.1 The Three-Layer Prediction Architecture



8.2 XGBoost — The Risk Scorer

****Algorithm****: Gradient-boosted decision trees (150 trees, max-depth 5, learning rate 0.05)

****Why XGBoost?****

- Handles mixed data types (continuous + categorical) naturally
- Robust to missing values
- Feature importance gives interpretable "driver" attribution
- Fast training and inference
- State-of-the-art for tabular data

****How features are extracted from 30-day window:****

```

30-day data window:
    soil_moisture_pct (D30)
    ndvi (D30)
    lst_c (D30)

    rainfall_7d_mm = sum(D24:D30)
    etc_7d_avg_mm = mean(D24:D30)
    soil_moisture_trend = linear fit(D24:D30)

```

```

ndvi_trend = linear_fit(D17:D30)

rainfall_30d_mm = sum(D1:D30)
etc_30d_avg_mm = mean(D1:D30)
temp_anomaly_c = D30_temp - mean(D1:D30)_temp

```

****The 11 engineered features:****

#	Feature	Formula	What It Detects
1	`soil_moisture_pct`	Latest value	Current water availability
2	`ndvi`	Latest value	Current vegetation health
3	`lst_c`	Latest value	Current surface heat stress
4	`rainfall_30d_mm`	Sum of 30 days	Medium-term water supply
5	`rainfall_7d_mm`	Sum of 7 days	Recent rainfall relief
6	`etc_7d_avg_mm`	Mean of 7 days	Recent water demand
7	`etc_30d_avg_mm`	Mean of 30 days	Sustained water demand
8	`temp_anomaly_c`	Latest - 30d mean	Heat wave detection
9	`days_since_last_rain`	Count of dry days	Drought duration
10	`soil_moisture_trend`	Linear slope over 7d	Dry-down rate
11	`ndvi_trend`	Linear slope over 14d	Vegetation decline rate

****Training target:****

```

soil_risk = (1 - min(1, soil_water / available_capacity)) × 100
ndvi_penalty = max(0, (0.8 - min(0.8, NDVI)) / 0.8) × 20
spi_penalty = max(0, -SPI) × 10
risk = min(100, soil_risk + ndvi_penalty + spi_penalty)

```

****Driver Attribution:****

XGBoost Feature	Maps to Driver Category
rainfall_30d_mm, rainfall_7d_mm, days_since_last_rain	**rainfall_deficit**
soil_moisture_pct, soil_moisture_trend	**soil_moisture_decline**
temp_anomaly_c	**high_temperature**
ndvi, ndvi_trend	**vegetation_stress**
etc_7d_avg_mm	**high_evapotranspiration**

Drivers are normalized to sum to 100%, providing a "what's causing this drought" breakdown.

8.3 LSTM — The Soil Moisture Forecaster

****Architecture:****

Input: (batch, 30-60 time steps, 10 features)

****10 input features to LSTM (the state vector):****

Index	Feature	Why Include
0	`soil_moisture_pct`	Target variable — the sequence we need to forecast
1	`rainfall_mm`	Primary driver that replenishes soil moisture
2	`temp_max_c`	Drives ET — drains soil moisture
3	`temp_min_c`	Affects nighttime respiration and dew formation
4	`humidity_pct`	Controls ET rate
5	`wind_speed_ms`	Affects ET rate
6	`solar_radiation_wm2`	Energy for ET
7	`ndvi`	Vegetation health — declines when stressed
8	`lst_c`	Surface heat — rises when dry
9	`etc_mm`	Actual water demand

****Why LSTM over other time-series models?****

- LSTMs capture long-term dependencies (up to 60 days in this case)
- They model non-linear relationships naturally
- They can learn complex interactions between variables (e.g., rain + temperature → soil moisture is non-linear)
- Outperforms ARIMA/SARIMA for multivariate environmental time series

****How training works:****

1. Normalize all 10 features to zero-mean unit-variance
2. Create sliding windows: for a dataset of N days, create (N - 30 - forecast_days) sequences
3. Each sequence: 30 days of [10 features] → target: soil moisture at day 30 + forecast_days
4. Split 80/20 train/validation
5. Train with Adam optimizer, MSELoss, early stopping (patience=15)
6. Two separate models:
 - ****LSTM-7d****: sequence=30, forecast=7 → predicts soil moisture 7 days ahead
 - ****LSTM-30d****: sequence=60, forecast=30 → predicts soil moisture 30 days ahead

****Converting LSTM soil moisture to risk:****

```
risk = (1 - min(1, predicted_soil_moisture / field_capacity)) × 100
```

This gives a 0-100 risk score. If predicted moisture is at or above field capacity, risk = 0. If moisture is 0, risk = 100.

8.4 Heuristic (Physics-Based) Fallback

When ML models are not trained, the system uses deterministic formulas:

****Risk calculation:****

```
soil_risk = (1 - min(1, current_moisture / field_capacity)) × 100
ndvi_penalty = max(0, (0.8 - ndvi) / 0.8) × 20
spi_penalty = max(0, -spi) × 10
risk_today = min(100, soil_risk + ndvi_penalty + spi_penalty)
```

****Future risk projection:****

```
decline_factor = max(0, min(2.0, abs(soil_trend) × 10))
rainfall_relief = max(0, min(20, rainfall_7d × 1.5))
risk_7day = max(0, min(100, risk_today + decline_factor × 8 - rainfall_relief × 0.5))
risk_30day = max(0, min(100, risk_today + decline_factor × 18 - rainfall_relief))
```

****Driver scores (heuristic):****

```
rainfall_deficit = max(0, 100 - min(100, (rainfall_30d/30) × 10)) + max(0, 100 - min(100, rainfall_7d × 10))
soil_moisture_decline = max(0, min(100, abs(soil_trend) × 300))
high_temperature = max(0, min(100, temp_anomaly × 15))
vegetation_stress = max(0, min(100, (1 - ndvi) × 100))
high_evapotranspiration = max(0, min(100, etc_avg × 20))
```

8.5 The Penman-Monteith ET0 Engine

A simplified FAO-56 Penman-Monteith is implemented in `physical\_helpers.py`:

```
es = 0.6108 × exp(17.27 × T / (T + 237.3))

ea = es × relative_humidity / 100

VPD = es - ea

Rn = solar_radiation_Wm2 × 0.0864 # convert to MJ/m2/day

# 7. Wind factor
wind_factor = 0.26 × (1 + 0.54 × wind_speed_ms)

# 8. Reference ET (FAO-56 Penman-Monteith)
```

8.6 The Daily Water Balance

The pipeline simulates the soil water tank:

```
soil_water[t] = soil_water[t-1] + rainfall[t] + irrigation[t] - ETc[t] - runoff[t]
```

Where:

- Initial condition: `soil\_water[0] = available\_water\_capacity × 0.7` (starting at 70% full)
- `runoff[t] = max(0, rainfall[t] - 5) × 0.3` (simple SCS-like runoff curve)
- `soil\_water[t]` can never go below 0

9. THE LLM EXPLAINER

The system optionally connects to a local ****Ollama**** server running ****Llama 3.2 (3B)**** to generate natural language explanations.

****Input prompt structure:****

```
You are a drought analysis assistant for Ichkeul Lake, Tunisia.
Generate a 2-3 sentence explanation for a farmer based on:
- Current drought risk: 65/100 (Severe)
- 7-day forecast: 72/100 (Severe)
- 30-day forecast: 80/100 (Extreme)
- Primary driver: 45% rainfall deficit
- Secondary driver: 30% temperature anomaly
- Soil moisture: 18% (35% of capacity)
```

****Fallback explanations** (when Ollama is unavailable):**

- Risk ≤ 20: "Conditions are normal..."
- Risk 21-40: "Watch level..."

- Risk 41-60: "Moderate drought..."
 - Risk 61-80: "Severe drought..."
 - Risk 81-100: "EXTREME DROUGHT EMERGENCY..."
-

10. END-TO-END PREDICTION FLOW

User clicks "Get Drought Prediction" (API: /api/drought-prediction/?region_id=X&year_id=Y)

1. Check database for recent prediction (within 1 hour)
 2. Initialize DroughtPredictionPipeline
 3. Set soil properties
 4. Build prediction frame (query last 120 days)
 5. Merge all on measurement_date (outer join)
 6. Prepare daily data
 7. Check for trained model files
 8. Predict
 9. Generate LLM explanation (optional)
 10. Save prediction to DroughtPrediction table
-

11. APPENDIX: FORMULA REFERENCE

Saxton & Rawls Pedotransfer Functions

Field Capacity (%) = $20.0 + 0.3 \times \text{clay} + 0.1 \times \text{silt} - 0.05 \times \text{sand}$
Wilting Point (%) = $8.0 + 0.2 \times \text{clay} + 0.05 \times \text{silt} - 0.03 \times \text{sand}$
Available Water Capacity (mm) = $(\text{FC} - \text{WP}) / 100 \times \text{root_depth_mm}$

Simplified Penman-Monteith (FAO-56)

$es = 0.6108 \times \exp(17.27 \times T / (T + 237.3))$
 $ea = es \times RH / 100$
 $VPD = es - ea$
 $Rn = \text{solar_radiation_Wm2} \times 0.0864$
 $\text{wind_factor} = 0.26 \times (1 + 0.54 \times u2)$

Crop Evapotranspiration

$ETc = ET0 \times Kc$

Daily Water Balance

$S_t = \max(0, S_{\{t-1\}} + P_t + I_t - ETc_t - R_t)$
 $R_t = \max(0, P_t - 5) \times 0.3$
 $S_0 = AWC \times 0.7$

Standardized Precipitation Index (SPI)

Clamped to [-3, +3]

Vegetation Condition Index (VCI)

$$\text{VCI} = (\text{NDVI}_{\text{current}} - \text{NDVI}_{\text{min}}) / (\text{NDVI}_{\text{max}} - \text{NDVI}_{\text{min}}) \times 100$$

Heuristic Drought Risk (Fallback)

$$\text{Soil Risk} = (1 - \min(1, \text{SM} / \text{FC})) \times 100$$
$$\text{NDVI Penalty} = \max(0, (0.8 - \text{NDVI}) / 0.8) \times 20$$
$$\text{SPI Penalty} = \max(0, -\text{SPI}) \times 10$$
$$\text{Total Risk} = \min(100, \text{Soil Risk} + \text{NDVI Penalty} + \text{SPI Penalty})$$

LSTM Risk Conversion

$$\text{Risk}_{\text{forecast}} = (1 - \min(1, \text{SM}_{\text{predicted}} / \text{FC})) \times 100$$
